

Standard Form & Indices

Test 9 · 41 marks total · Sections F & G include Higher-tier content

Name: _____

Date: _____

Score: _____ / 41

Section A — Laws of indices

A1. Write $8^5 \times 8^4$ as a single power of 8.

[1 mark]

A2. Write $7^{14} \div 7^6$ as a single power of 7.

[1 mark]

A3. Write $(2^6)^3$ as a single power of 2.

[1 mark]

A4. Work out the value of $2^3 \times 2^3$.

[2 marks]

Section B — Zero and negative indices

B1. Work out the value of 350^0 .

[1 mark]

B2. Work out the value of 12^{-2} . Give your answer as a fraction.

[2 marks]

B3. Work out the value of 20^{-1} . Give your answer as a decimal.

[2 marks]

Section C — Writing standard form

C1. Write 18,000 in standard form.

[2 marks]

C2. Write 3,900,000 in standard form.

[2 marks]

C3. Write 0.0056 in standard form.

[2 marks]

C4. Write 0.00002 in standard form.

[2 marks]

Section D — Converting back, and calculating

D1. Write 9.4×10^4 as an ordinary number.

[1 mark]

D2. Write 3×10^{-6} as an ordinary number.

[1 mark]

D3. Work out $(8 \times 10^4) \times (5 \times 10^3)$. Give your answer in standard form.

[3 marks]

D4. Work out $(5 \times 10^7) \div (2 \times 10^3)$. Give your answer in standard form.

[3 marks]

Section E — Powers to know, and estimating roots

E1. Write down the value of 6^3 .

[1 mark]

E2. $\sqrt{30}$ lies between which two consecutive whole numbers?

[1 mark]

Section F — Laws of indices with algebra

F1. Simplify $c^4 \times c^{11}$.

[1 mark]

F2. Simplify $k^{18} \div k^5$.

[1 mark]

F3. Simplify $(m^3)^6$.

[1 mark]

F4. Simplify $5p^4 \times 6p^2$.

[2 marks]

F5. Simplify $(2q^3)^5$.

[2 marks]

Section G — Fractional indices (Higher)

G1. Work out the value of $169^{1/2}$.

[1 mark]

G2. Work out the value of $729^{1/3}$.

[1 mark]

G3. Work out the value of $8^{4/3}$.

[2 marks]

G4. Work out the value of $100^{-1/2}$.

[2 marks]